Consumer Learning and Competitive Inefficiency

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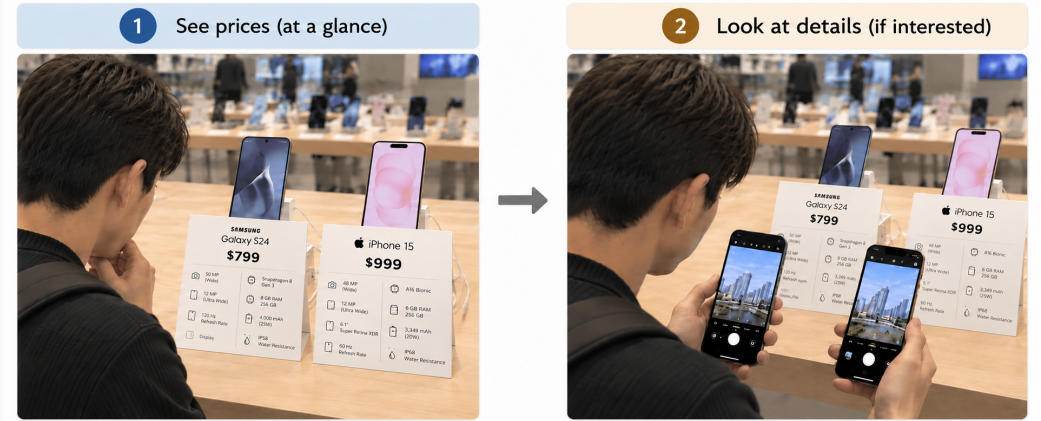
Motivation

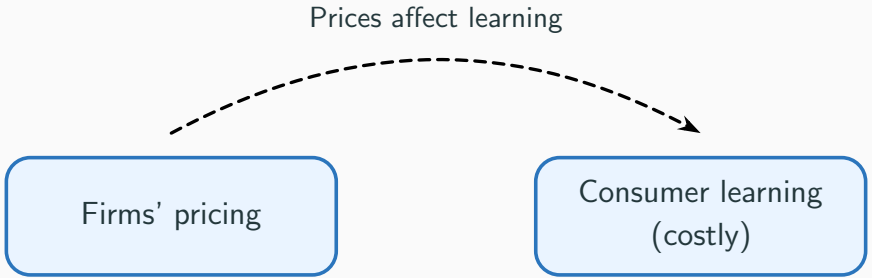
- Consumers often observe prices before learning which product fits them better.
- **Conventional wisdom:** with easier access to product information, consumers make better choices and are better off.

Main Question

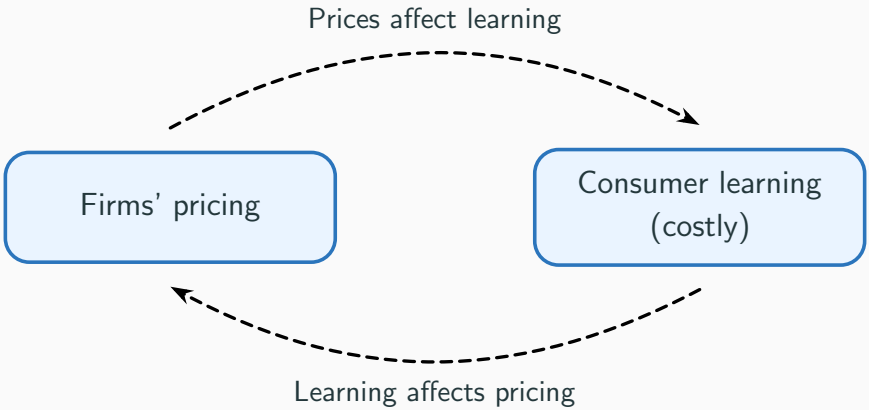
Does easier access to product information necessarily improve welfare in competitive markets?

Would you compare the phones, or simply buy the cheaper one?





- Value of learning depends on the observed prices.



- Value of learning depends on the observed prices.
- Yet learning shapes how firms compete.

What We Do

This paper studies the interaction between firms' pricing decisions and consumer learning.

We characterize the **unique mixed-strategy pricing equilibrium** when both firms are active.

Key Findings:

- Even when information is free, price competition generates inefficient product choices.
- Higher information costs can intensify price competition and **raise consumer surplus**.
- Depending on market asymmetry, **either free information or no learning** maximizes total surplus.

Endogenous Transparency: Firms may provide more transparency than is socially optimal.

Roadmap

1. Model
2. Equilibrium Characterization
3. Welfare Analysis
4. Quasi-Monopoly
5. Endogenous Transparency

Model: Environment

- Two firms produce differentiated goods at zero marginal cost and choose prices $p_i \in [0, \bar{p}]$, $i \in \{1, 2\}$.
- A unit-demand consumer has valuation v_i for Firm i's product.
- Let $w := v_1 - v_2$ denote the relative match value, and write

$$(v_1, v_2) = \left(V + \frac{w}{2}, V - \frac{w}{2}\right).$$

- Purchasing from Firm i yields utility $v_i - p_i$. We assume $V > \bar{p}$ so that market is covered.

Model: Consumer Learning

- The relative match value is binary: $w = s > 0$ with probability $\mu \in (0, 1)$, and $w = -1$ with probability $1 - \mu$.
  - μ determines how **often** Firm 1 is the better match, while s determines the **size** of its advantage.
- The prior mean is $\bar{w} := \mu s - (1 - \mu)$.
  - **weak ex ante advantage**: Assume $\bar{w} \geq 0$.
  - symmetry: $(\mu, s) = (1/2, 1)$; $\bar{w} = 0$ need not imply symmetry.
- The consumer does not initially observe w but can acquire information that reveals it.
- The cost of learning is

$$C(k) := k \operatorname{Var}(w),$$

  where $k \geq 0$ is the unit information cost.

Timing

1. The firms simultaneously choose prices.
2. After observing both prices, the consumer decides whether to acquire information about w.
3. The match value w is realized. The consumer observes it if she chose to acquire information; otherwise, she remains uninformed.
4. The consumer makes her purchase decision.

Solution concept: Subgame perfect equilibrium (SPE).

Related Literature

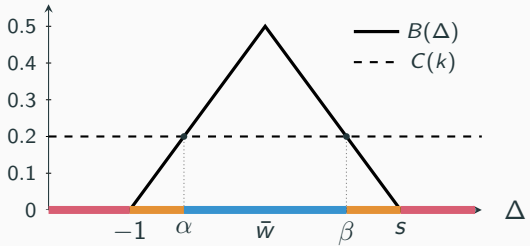
- **Costly learning**: Branco, Sun, and Villas-Boas (2012), **Ravid, Roesler, and Szentes (2022)**, Biglaiser, Gu, and Li (2025), **Albrecht and Whitmeyer (2026)**, Mensch, Ravid, and Thereze (2026).
- **Consumer information and pricing**: Moscarini and Ottaviani (2001), Roesler and Szentes (2017), Armstrong and Zhou (2022), Elliott et al. (2026).
- **Price dispersion**: Shilony (1977), Rosenthal (1980), Varian (1980), Narasimhan (1988), Raju, Srinivasan, and Lal (1990), Jing and Wen (2008), Armstrong and Vickers (2019, 2022).
- **Consumer search**: Wolinsky (1986), Stahl (1989), Anderson and Renault (1999), **Armstrong and Zhou (2011)**, **Choi, Dai, and Kim (2018)**, **Haan, Moraga-González, and Petrikaitė (2018)**, Jain and Whitmeyer (2021), Chen et al. (2021), **Obradovits and Plaickner (2023)**.
- **Information provision and transparency**: Anderson and Renault (2009), Bar-Isaac, Caruana, and Cuñat (2010), Gu and Xie (2013), Board and Lu (2018), Au and Whitmeyer (2023), Hwang, Kim, and Boleslavsky (2026), Jain and Whitmeyer (2026).

Roadmap

1. Model
2. **Equilibrium Characterization**
3. Welfare Analysis
4. Quasi-Monopoly
5. Endogenous Transparency

Consumer's Problem

- Let $\Delta := p_1 - p_2$. The consumer's indirect utility is
$$u(w) := \max\left\{V + \frac{w}{2} - p_1,\ V - \frac{w}{2} - p_2\right\}.$$
- If informed, she chooses based on w; otherwise, on $\bar{w}$.
- The benefit of learning is $B(\Delta) := \mu u(s) + (1-\mu)u(-1) - u(\bar{w})$.
- It is maximized at $\Delta = \bar{w}$: the consumer is most uncertain.
- $B(\Delta) = C(k)$ yields two cutoffs $\alpha := -1 + k\mu(1+s)^2$ and $\beta := s - k(1-\mu)(1+s)^2$.



Demand

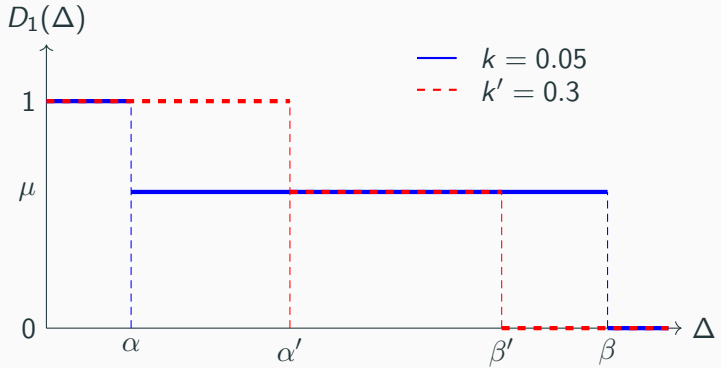
- The consumer learns if and only if $\Delta \in [\alpha, \beta]$.
  - assume $k < \bar{k} := 1/(1+s)$ so that $\beta - \alpha > 0$.
- The consumer's optimal learning and purchase decisions yield

$$D_1(\Delta) = \begin{cases} 1, & \Delta < \alpha, \\ \mu, & \alpha < \Delta < \beta, \\ 0, & \Delta > \beta, \end{cases} \qquad D_2(\Delta) = 1 - D_1(\Delta)$$

where $D_1(\alpha) = d_\alpha \in [\mu, 1]$ and $D_1(\beta) = d_\beta \in [0, \mu]$ are determined by admissible tie-breaking rules.

Demand

- **Reduced differentiation:** as k increases, the learning region $L := \beta - \alpha$ shrinks.



$(\mu, s) = (0.6, 1)$.

No Pure Strategy Equilibrium

- Define
$$\hat{k} := \frac{1+\mu s}{\mu(2-\mu)(1+s)^2} \leq \bar{k}.$$
- Assume $\bar{p} > 3 + 4s$.

Lemma 1. For $0 \leq k < \hat{k}$, no pure-strategy pricing equilibrium exists under any admissible tie-breaking rule.

1. Any pure equilibrium must satisfy $\Delta \in \{\alpha, \beta\}$.
2. $\Delta = \beta$ cannot be sustained:
   - If $d_\beta < \mu$, Firm 1 slightly lowers its price.
   - If $d_\beta = \mu$, Firm 2 slightly raises its price.
3. At $\Delta = \alpha$, profitable deviations leave only $(p_1, p_2) = (\alpha, 0)$ with $d_\alpha = 1$.
4. However,
$$\alpha - \mu\beta = \mu(2-\mu)(1+s)^2(k-\hat{k}) < 0.$$
Firm 1 can profitably deviate to $p_1 = \beta - \varepsilon$: it prefers sharing the market to capturing it at a lower price.

Mixed Strategy Equilibrium

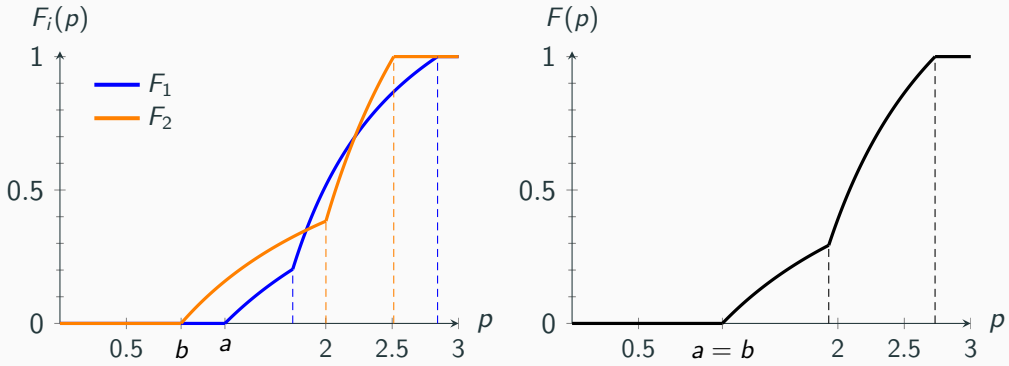
Proposition 1. For $0 \leq k < \hat{k}$, there is a unique equilibrium in price distributions under any admissible tie-breaking rule. The equilibrium price distributions F_1 and F_2 are atomless, and there exist $a, b > 0$ such that

$$\text{supp}(F_1) = [a, a + L], \qquad \text{supp}(F_2) = [b, b + L],$$

where $a < b + \beta < a + L$ and $b < a - \alpha < b + L$.

- **Main challenge:** Pin down the equilibrium support structure under asymmetry.
  - Rule out atoms.
  - Show that both supports have length $L = \beta - \alpha$ and are connected.
  - Indifference and continuity conditions uniquely determine the distributions.
  - Verify that the distributions are mutual best responses.

Mixed Strategy Equilibrium



Left: Asymmetric $(\mu, s, k) = (0.6, 1, 0.1)$; Right: Symmetric $(\mu, s, k) = (0.5, 1, 0.1)$.

Mixed Strategy Equilibrium

- The **switching prices** $b+\beta$ and $a-\alpha$ divide each support into lower ($\mathcal{L}_i$) and upper branches ($\mathcal{U}_i$).
  - At each switching price, the consumer always learns regardless of the rival's price.
  - **Two competing incentives**: $p_i \in \mathcal{L}_i$ guarantees positive demand, while $p_i \in \mathcal{U}_i$ offers higher margins but risks zero demand.
- Firm 1's expected profit against F_2 is

$$\pi_1(p) = p\big[1-(1-\mu)F_2(p-\alpha)-\mu F_2(p-\beta)\big].$$

- **Indifference:** Every price on the support yields the same profit. For Firm 1,

$$\Pi_1 = \begin{cases} p[1-(1-\mu)F_2(p-\alpha)], & p \in \mathcal{L}_1, \\ \mu p[1-F_2(p-\beta)], & p \in \mathcal{U}_1. \end{cases}$$

At $p = b+\beta$, we have $\Pi_1 = \mu(b+\beta)$.

Mixed Strategy Equilibrium

- **Equilibrium CDFs** (on their supports; $F_i = 0$ below and $F_i = 1$ above):

$$F_1(p) = \begin{cases} 1 - \dfrac{a-\alpha}{p-\alpha}, & p \in \mathcal{L}_1, \\ \dfrac{1}{\mu}\left(1 - \dfrac{(1-\mu)(a-\alpha)}{p-\beta}\right), & p \in \mathcal{U}_1, \end{cases} \qquad F_2(p) = \begin{cases} 1 - \dfrac{b+\beta}{p+\beta}, & p \in \mathcal{L}_2, \\ \dfrac{1}{1-\mu}\left(1 - \dfrac{\mu(b+\beta)}{p+\alpha}\right), & p \in \mathcal{U}_2. \end{cases}$$

- **Endpoint equations:** Continuity at the switching prices determines (a, b):

$$1 - \frac{b+\beta}{a+L} = \frac{1-\mu(b+\beta)/a}{1-\mu}, \qquad 1 - \frac{a-\alpha}{b+L} = \frac{1-(1-\mu)(a-\alpha)/b}{\mu}.$$

- Equilibrium profits are $\Pi_1 = \mu(b+\beta)$ and $\Pi_2 = (1-\mu)(a-\alpha)$.
- Henceforth, we focus on $k < \hat{k}$.

Lemma 2. For $0 \le k_1 < k_2$, $F_i(p; k_2) \ge F_i(p; k_1)$ for all $p \in \mathbb{R}$, $i = 1, 2$.

- Higher information costs reduce product differentiation and intensify price competition.

Roadmap

1. Model
2. Equilibrium Characterization
3. **Welfare Analysis**
4. Quasi-Monopoly
5. Endogenous Transparency

Failure of Ex Post Efficiency

- **First best:** With costless information, the consumer is assigned to the better-matched product in each state:

$$TS^{FB} = V + \frac{\mu s + (1-\mu)}{2}.$$

- **Equilibrium total surplus:** Let $\Delta = p_1 - p_2$ denote the (random) equilibrium price gap. Then

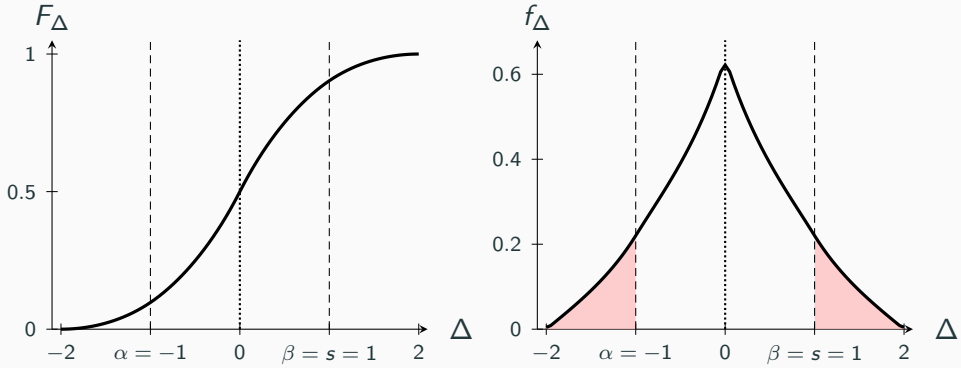
$$TS^{FB} - TS(k) = \underbrace{(1-\mu)\Pr(\Delta < \alpha) + \mu s \Pr(\Delta > \beta)}_{\text{Misallocation}} + \underbrace{C(k)\Pr(\alpha \le \Delta \le \beta)}_{\text{Learning cost}}.$$

Proposition 2. In equilibrium,

$$[\alpha, \beta] \subset \text{int}(\text{supp}(\Delta)).$$

- Price dispersion can induce the consumer to choose the worse-matched product.
- Extreme asymmetry: As $\mu \to 1$, total surplus converges to the first best.

Free Information Benchmark



Symmetric case: $(\mu, s, k) = (1/2, 1, 0)$. Shaded tails occur with probability around 19.4%.

Information Costs and Welfare: $\bar{w} = 0$

Lemma 3. Suppose $\bar{w} = 0$ and let $\lambda(k) = 1 - k(1+s)$. Then,

$$F_i(p;k) = F_i\left(\frac{p}{\lambda(k)};0\right), \quad \Delta(k) \overset{d}{=} \lambda(k)\Delta(0), \quad \Pi_i(k) = \lambda(k)\Pi_i(0).$$

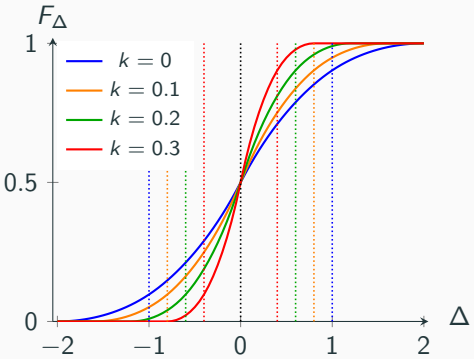
- The learning region ($\alpha = -\lambda$ and $\beta = s\lambda$) and equilibrium price gap shrink by the same factor λ.
- Hence, the probability of learning (and misallocation probabilities) remains constant for $k > 0$:

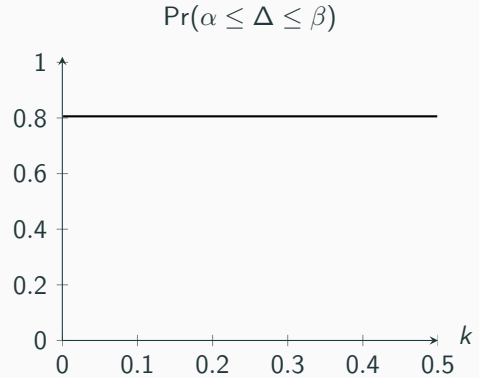
$$P_0 := \Pr(-1 \le \Delta(0) \le s) = \Pr(\alpha \le \Delta(k) \le \beta).$$

- **Total surplus:** Since $C(k) = sk$,

$$TS(k) = TS(0) - \underbrace{skP_0}_{\text{Learning cost}}.$$

- Higher k does not change the allocation but only reduces total surplus.


F_Δ
1
0.5
0
$k = 0$
$k = 0.1$
$k = 0.2$
$k = 0.3$
−2
−1
0
1
2
Δ


$\Pr(\alpha \leq \Delta \leq \beta)$
1
0.8
0.6
0.4
0.2
0
0
0.1
0.2
0.3
0.4
0.5
k

$(\mu, s) = (0.5, 1)$.

Information Costs and Allocation: $\bar{w} > 0$

Lemma 4. Suppose $\bar{w} > 0$. As $k \uparrow \hat{k}$,

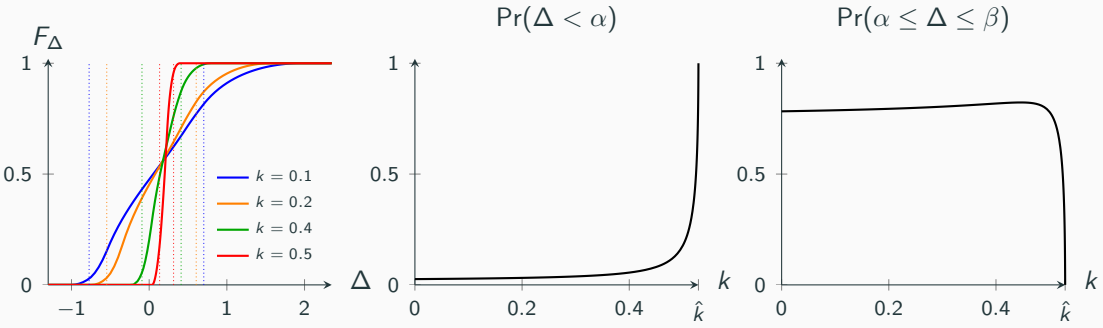
$$\Pr(p_1 \in \mathcal{L}_1; k) \uparrow 1 \quad \text{and} \quad \Pr(p_2 \in \mathcal{U}_2; k) \uparrow 1.$$

- Firm 1: Capturing the entire market becomes easier.
- Firm 2: Firm 1's aggressive pricing makes price cuts less attractive.

Implications: As $k \uparrow \hat{k}$,

- $\Pr(\Delta < \alpha) \uparrow 1$ and $\Pr(\Delta > \beta) \downarrow 0$.
- Equilibrium demands $\bar{D}_1(k) \uparrow 1$ and $\bar{D}_2(k) \downarrow 0$.

Information Costs and Allocation: $\bar{w} > 0$



$(\mu, s) = (0.7, 0.8)$.

Information Costs and Welfare: $\bar{w} > 0$

- Let $P(k) := \Pr(\alpha \leq \Delta \leq \beta)$.
- Let $X = 1$ if the consumer buys from Firm 1 and $X = 0$ otherwise.

$$TS(k) = \mathbb{E}\left[V + \frac{(2X-1)w}{2}\right] - C(k)P(k)$$

$$= V + \underbrace{\bar{w}\left(\bar{D}_1(k) - \frac{1}{2}\right)}_{\text{Allocation effect}} + \underbrace{\text{Cov}(X, w)}_{\text{Matching effect}} - \underbrace{C(k)P(k)}_{\text{Learning cost}}$$

  where $\text{Cov}(X, w) = \mu(1-\mu)(1+s)P(k)$.

- As k increases, demand shifts toward the advantaged firm, raising the **allocation effect**.
- **Matching effect:** Learning helps the consumer choose the better-matched product in each state:

$$\mathbb{E}[\max\{v_1, v_2\}] - \left[\mu\mathbb{E}[v_1] + (1-\mu)\mathbb{E}[v_2]\right] = \mu(1-\mu)(1+s) > 0.$$

- Thus, the effect of k on total surplus is **ambiguous**.

Information Costs and Welfare: $\bar{w} > 0$

Proposition 3. Fix $s > 0$. There exists a unique cutoff $\mu^*(s) \in (1/(1+s), 1)$ such that

$$TS(k) \text{ is maximized } \begin{cases} \text{at } k = 0 & \text{if } \mu \le \mu^*(s), \\ \text{at } k = \hat{k} & \text{if } \mu \ge \mu^*(s). \end{cases}$$

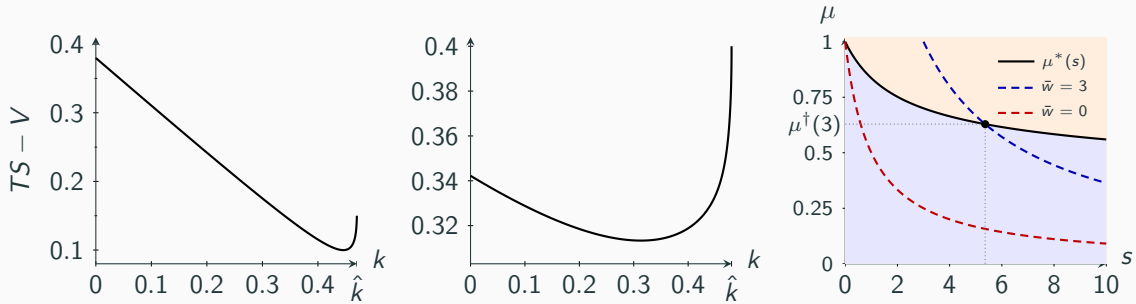
- No information cost is incurred: at $k = 0$, information is free; at $k = \hat{k}$, the consumer never learns.
- Comparing the extremes,

$$\begin{aligned} TS(0) - TS(\hat{k}) &= \underbrace{\mu(1-\mu)(1+s)P_0}_{\text{Matching gain}} - \underbrace{\bar{w}[1 - \bar{D}_1(0)]}_{\text{Allocation loss}} \\ &= (1-\mu)P_0 - \bar{w}R_0, \end{aligned}$$

  where $P_0 = \Pr(-1 \le \Delta(0) \le s)$ and $R_0 = \Pr(\Delta(0) > s)$.
- Higher μ: Less gain from learning, while buying from Firm 2 regardless of fit becomes more costly.
- **Why no interior optimum?:** An interior k incurs learning costs without sufficient matching or allocation gains.
- Moreover, $\mu^*(s) \downarrow 0$ as $s \to \infty$: sufficiently large s makes no learning optimal for any fixed μ.

Information Costs and Welfare: $\bar{w} > 0$

Corollary 1. Fix $\bar{w} > 0$ and vary (μ, s) with $\mu s - (1-\mu) = \bar{w}$. There exists a unique cutoff $\mu^\dagger(\bar{w}) \in (0,1)$ such that $TS(k)$ is maximized at $k = 0$ if $\mu < \mu^\dagger(\bar{w})$, and at $k = \hat{k}$ if $\mu \geq \mu^\dagger(\bar{w})$.



$(\mu, s) = (0.65, 1)$ (left), $(0.9, 1)$ (middle); $\mu^*(1) \simeq 0.836$.

Consumer Surplus

- Consumer surplus is
$$CS(k) = TS(k) - \Pi_1(k) - \Pi_2(k).$$

Proposition 4. Consumer surplus is strictly increasing in k.

- At given prices, the consumer's payoff is
$$\max\left\{\mathbb{E}[v_1] - p_1,\ \ \mathbb{E}[v_2] - p_2,\ \ \mathbb{E}[\max\{v_1, v_2\}] - \mu p_1 - (1-\mu)p_2 - C(k)\right\}.$$
- **Key property:** For every quantile $u \in (0, 1)$,
$$Q_i(u, k) + C(k) \quad \text{strictly decreases in } k, \quad i = 1, 2.$$
- Thus, at any fixed pair of price quantiles, all three payoffs increase.

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4. **Quasi-Monopoly**
5. Endogenous Transparency

Quasi-Monopoly

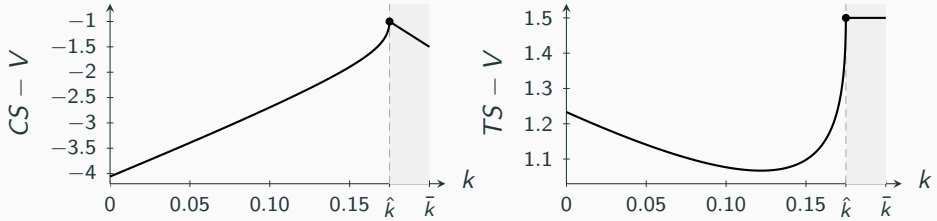
Proposition 5. Suppose $\bar{w} > 0$ and $k \in [\hat{k}, \bar{k})$. Under any admissible tie-breaking rule, there is an equilibrium in which Firm 1 sets $p_1 = \alpha$ and Firm 2 mixes according to

$$F_2(p) = \frac{p}{(1-\mu)(\alpha+p)}, \quad p \in [0, L), \qquad F_2(L) = 1.$$

The consumer does not acquire information and purchases from Firm 1 almost surely. The equilibrium outcomes are $(\Pi_1, \Pi_2) = (\alpha, 0)$, $TS = \mathbb{E}[v_1]$, $CS = \mathbb{E}[v_1] - \alpha$.

- Price strategies need not be unique, but the equilibrium outcome is unique.
- Although Firm 2 earns zero profit, the threat of undercutting and the possibility of consumer learning constrain Firm 1's price: $\alpha < \bar{w}$.
- Higher information costs allow Firm 1 to raise its price without inducing learning, reducing consumer surplus.

Quasi-Monopoly



$(\mu, s) = (0.8, 4)$. Shading indicates the quasi-monopoly regime.

Lemma 5. For $k \in [0, \bar{k})$, Firm 1's profit is uniquely maximized at $k = 0$.

- Thus, Firm 1 itself prefers the competitive market under free information to any quasi-monopoly outcome because consumer learning softens price competition.

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Endogenous Transparency

- So far, information costs have been exogenous. In practice, firms can influence how easily consumers compare products (e.g., T-Mobile and Gladia).
- Greater transparency can improve product matching but also soften price competition. Would the advantaged firm choose the optimal level of transparency for consumers or society?
- Firm 1 chooses $k \in [0, K]$, with $K \geq 1$, before price competition. Lower k means greater transparency, with provision cost $c(k)$ satisfying $c'(k) < 0 < c''(k)$.
- Let k^f, k^c, and k^w denote the choices maximizing Firm 1's net profit, consumer surplus, and net total surplus, respectively.

Proposition 6. Firm 1's optimal transparency choice k^f is generically unique. If $k^f = K$, then $k^w = K$.

Endogenous Transparency

- **Example** ($\bar{w} = 0$). For $k \leq \bar{k}$,

$$\Pi_1(k) = \lambda(k)\Pi_1(0), \quad TS(k) = TS(0) - ksP_0.$$

- In this case, $k^f \leq k^w$: Firm 1's marginal gain from transparency exceeds the social gain:

$$(1+s)\Pi_1(0) = b_0 + s > sP_0.$$

Lemma 6.

1. For fixed $s > 0$ and μ sufficiently close to one, $k^c = \hat{k} < k^f = k^w = K$.
2. For fixed $\mu \in (0,1)$ and sufficiently large s, $k^f = 0 < k^c = \hat{k} < k^w = K$.

- When μ is close to one, learning adds little value to Firm 1: $\Pi_1(K) = \bar{w}$ and $\Pi_1(0) = \mu(b_0 + s)$ both converge to s. Firm 1 thus saves on transparency costs.
- When s is sufficiently large, learning supports higher prices, making transparency valuable.

Conclusion

Main takeaway.

Easier access to information need not improve market outcomes: learning also changes firms' pricing incentives.

- Even with free information, price competition can induce consumers to choose the worse-matched product.
- Learning softens price competition. Thus, higher information costs can benefit consumers.
- Higher information costs can also shift market allocation toward the advantaged firm. Their effect on total surplus depends on market asymmetry.

Policy implications.

- Firms may provide more transparency than is socially optimal. Mandating greater transparency when firms choose opacity can reduce net total surplus.
- Transparency policy should account for firms' pricing incentives, market asymmetry, and the policy objective.